

Biju Patnaik University of Technology, Odisha, Rourkela
FDP on “Artificial Intelligence and Machine Learning”

Conceptual Foundations of Machine Learning: A Hands-On Session

By Rashmikiranjan Nayak on 09.02.2026, (9.00 AM -11.00 AM)

Key Takeaway

1. Overparameterization & Generalization

Generalization is determined by solution selection. Optimization implicitly prefers stable, low-complexity solutions rather than arbitrary fits.

2. Training Loss $\neq$ Learning

Perfect training accuracy does not imply learning. True learning must be validated through generalization on unseen data.

3. Inductive Bias

Learning is impossible without inductive bias. Bias arises from the model class, regularization, optimization strategy, and data representation.

4. Simple vs Complex Models

Simple models often outperform complex ones in small or noisy data regimes due to lower variance and greater stability.

5. Regularization

Regularization selects stable solutions among many that fit the training data equally well. It primarily controls solution geometry and robustness.

6. Accuracy vs Model Quality

Accuracy ignores uncertainty and confidence. Well-calibrated models with slightly lower accuracy may be superior for real-world decisions.

7. Bias–Variance Limitation

Classical bias–variance theory fails in the interpolation regime. Double descent explains why highly complex models can still generalize.

8. Statistical Significance

Single-run results are anecdotal. Reliable evaluation requires reporting the mean $\pm$ variance across multiple runs and performing statistical testing.

9. Representation vs Classifier

Good representations make learning easy. With the right representation, even simple classifiers can perform well.

10. Dataset Bias & Sanity Checks

Trust models that fail when the true signal is removed. Use label shuffling, feature ablation, and stress tests to detect shortcut learning.

Key Message

Machine learning is not about algorithms alone. It is about reasoning under uncertainty with rigor, proper validation discipline, and scientific skepticism.